

Comparison of YOLOv8s, YOLO11s, and YOLOv8s+CBAM for Motorcyclist Helmet-Use Detection

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Abstract—Helmet use is the primary protective factor for motorcyclists, and automatic compliance detection from traffic cameras can support road-safety enforcement. This study compares three YOLO-family architectures for helmet-use detection: YOLOv8s as a pure-CNN baseline, YOLO11s with partial self-attention through C2PSA blocks, and YOLOv8s augmented with CBAM channel-spatial attention. We hypothesize that YOLO11s, as the more recent architecture, will outperform the plain baseline. All models were trained on a public dataset of 1,803 images under an identical protocol and evaluated on a held-out test split with multiple seeds and paired t-tests. The results do not confirm the hypothesis. YOLOv8s achieved the highest mAP@0.5 (0.9592 ± 0.0014) at approximately 296 FPS. YOLO11s did not differ significantly from the baseline ($p = 0.54$). Adding CBAM significantly reduced mAP@0.5 by 0.0113 ($t(2) = 10.9$, $p = 0.008$, Cohen’s $d \approx 9.1$). Within this scope, added attention complexity does not improve accuracy on a small, relatively easy helmet-detection dataset.

Index Terms—helmet detection, object detection, attention mechanism, YOLOv8, YOLO11, CBAM, motorcycle safety

I. INTRODUCTION

Motorcycles account for a large share of traffic-accident casualties, especially in countries where they dominate daily mobility. The helmet is the most decisive protective factor against fatal head injuries, and monitoring compliance at scale requires automation. Manual observation does not scale to real traffic volumes, whereas automatic object detection from camera footage can operate continuously and consistently [1].

The YOLO family has become the de facto standard for real-time detection, with each iteration improving the backbone, neck, and label-assignment strategy [2]–[5]. YOLO11 is the most recent iteration and introduces positional self-attention blocks (C2PSA) into the feature pathway, making it a natural successor to the attention-free YOLOv8. Separately, lightweight attention modules such as CBAM [6] or Squeeze-and-Excitation (SE) networks [7] are a popular way to inject channel-spatial attention into an existing CNN backbone without replacing the architecture entirely.

Several recent studies inject attention modules into YOLO-family detectors and report accuracy improvements on helmet-detection datasets [8]–[10]. These reports create an expectation

that adding attention should help. However, the studies generally compare a single configuration without repetition across seeds and omit statistical significance tests. Consequently, the reported gains could stem from hyperparameters, augmentation, or initialisation rather than from the attention mechanism itself.

We address this gap by evaluating three architectures under a unified protocol. We hypothesize that YOLO11s, as the more recent architecture incorporating C2PSA blocks, will outperform the plain YOLOv8s baseline. The research question is: **does adding attention to a YOLO-family detector improve helmet-detection accuracy in a controlled, multi-seed comparison?**

The contributions of this paper are threefold. First, we present a controlled comparison of YOLOv8s, YOLO11s, and YOLOv8s+CBAM with identical hyperparameters, data, and augmentation, so that performance differences can be attributed to architectural factors. Second, we conduct a multi-seed evaluation that reports paired t-tests, Cohen’s d effect sizes, and 95% confidence intervals, so the conclusions do not rest on the luck of a single initialisation. Third, we provide a principled analysis of why the hypothesis was not confirmed by the data, along with practical implications for model selection in the helmet-detection domain.

II. LITERATURE REVIEW

A. Helmet Detection

Research on helmet detection has evolved from simple classification toward pipelines that detect, track, and classify motorcycles together with their riders. Siebert and Lin [1] demonstrated large-scale helmet-use detection from traffic video using a single-stage detector and released a widely used annotated dataset. In the enforcement context, several pipelines combine helmet detection with license-plate localisation and recognition to identify violators [8], [11], [12].

A growing body of work integrates attention mechanisms into helmet-detection models. Common patterns include adding channel-spatial attention modules to YOLO-family detectors and inserting feature-fusion blocks together with attention layers [9], [10]. These studies typically report accuracy

improvements over their respective baselines. However, the studies generally compare a single configuration without repetition across seeds, use protocols tuned per model, and omit statistical significance tests. This methodological gap directly motivates the controlled, multi-seed comparison presented in this study.

B. YOLO Architectures and Attention Modules

Single-stage detectors formulate detection as direct regression of bounding boxes and classes in a single network pass, trading some accuracy for high speed and employing specialized loss functions like Focal Loss [13] to mitigate class imbalance [2]. The YOLO family has become the de facto standard for real-time detection [3]. YOLOv8 represents a mature CNN generation without explicit attention modules, relying on efficient convolutional blocks, Feature Pyramid Networks (FPN) [14], and Path Aggregation Networks (PANet) [15] to handle multi-scale objects. YOLO11 introduces C2PSA, a positional self-attention block embedded into the feature pathway, which makes it a useful intermediate point between a pure CNN and a full transformer design.

Surveys of attention in vision [16], [17] conclude that the effectiveness of attention mechanisms depends heavily on training-data volume and task complexity. On the lightweight side, modules like CBAM [6] and Coordinate Attention [18] sequentially apply channel and spatial attention to emphasise both what is relevant and where it appears in a feature map. They add a small number of parameters and can be inserted into any CNN without replacing the backbone. For small-object detection, Cheng et al. [19] observe that objects such as helmets and license plates within wide frames require multi-scale feature representations and high input resolution.

III. METHODOLOGY

A. Dataset

We use a public helmet-detection dataset from Roboflow Universe (NCKH 2023 Helmet Detection Project, version 19, MIT license). The dataset contains 1,803 annotated images in YOLO format with three classes: helmet, license plate, and motorcyclist. A fixed split of 1,563 training, 140 validation, and 100 test images was used throughout all experiments. All final metrics were computed on the test split, which was never seen during training.

B. Architectures Compared

Three architectures were chosen to occupy different positions on the attention spectrum, as summarised in Table I. YOLOv8s represents the pure-CNN baseline with no explicit attention. YOLO11s adds partial self-attention through C2PSA blocks while remaining a single-stage detector. YOLOv8s+CBAM applies a controlled attention intervention by inserting CBAM modules into the baseline network.

Three CBAM modules were inserted at the outputs of the three detection scales (P3, P4, P5) just before the detection head, so the YOLOv8s+CBAM variant is identical to the plain YOLOv8s except for the added attention. The addition raises

TABLE I
THE THREE ARCHITECTURES AND THEIR ATTENTION MECHANISM.

Model	Attention	Params
YOLOv8s	None (baseline)	~11.17 M
YOLO11s	Partial (C2PSA)	~9.4 M
YOLOv8s+CBAM	Channel + spatial	~11.51 M

the parameter count from 11.17 to 11.51 million, an increase of approximately three percent. This small increment isolates the effect of attention while keeping all other factors constant.

C. Training Protocol

All models were trained with an identical protocol using the Ultralytics framework [20] on PyTorch with a single NVIDIA RTX 4090 GPU. Each model was initialised with COCO-pretrained weights [21]. For the CBAM variant, the attention layers were randomly initialised while the rest of the network inherited COCO weights. The unified configuration used automatic batch-size selection, automatic optimizer selection, identical augmentation (mosaic, HSV shifting, horizontal flipping), and early stopping with patience 25 over a maximum of 100 epochs. All random sources were seeded for reproducibility.

D. Training Configuration Rationale

The input resolution was set to 1280 pixels rather than the 640-pixel Ultralytics default. This choice was motivated by the small relative size of the helmet and license-plate classes within each frame. Inspection of the training images confirmed that these objects occupy a small fraction of the total pixel area, and higher resolution preserves the spatial detail needed for accurate bounding-box localisation. This is consistent with findings from Cheng et al. [19], who observe that high input resolution is critical for detecting small objects.

E. Evaluation Metrics and Statistical Testing

The primary metrics are mean Average Precision (mAP) at intersection-over-union (IoU) thresholds of 0.5 (mAP@0.5) and over a range of thresholds from 0.5 to 0.95 (mAP@[.5:.95]), supplemented by inference speed in frames per second (FPS). Average Precision (AP) for a given class is defined as the area under the precision-recall curve:

$$AP = \int_0^1 P(R) dR \quad (1)$$

where $P(R)$ denotes precision as a function of recall R . The mAP is subsequently computed as the arithmetic mean of AP across the set of target classes C :

$$mAP = \frac{1}{|C|} \sum_{c \in C} AP_c \quad (2)$$

FPS was measured on a single dedicated run while the GPU was idle, using the Ultralytics built-in benchmark on the full test split, with batch size 1 on the same hardware for all models.

TABLE II
PERFORMANCE ON THE TEST SPLIT (MEAN $\pm$ STD, $n = 3$ SEEDS).

Model	mAP@0.5	mAP@.5:95	FPS
YOLOv8s	0.9592 $\pm$ 0.0014	0.6862 $\pm$ 0.0046	$\sim$ 296
YOLO11s	0.9569 $\pm$ 0.0046	0.6898 $\pm$ 0.0030	$\sim$ 189
YOLOv8s+CBAM	0.9479 $\pm$ 0.0004	0.6810 $\pm$ 0.0045	$\sim$ 290

To assess the significance of accuracy differences, we paired the results between models on the same seed and applied a paired t-test to the mAP@0.5 difference of each seed pair, with significance level $\alpha = 0.05$. The paired t -statistic is defined as:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \quad (3)$$

where $\bar{d}$ represents the sample mean of the paired differences, s_d is the sample standard deviation of these differences, and n is the number of pairs ($n = 3$ seeds). To quantify the magnitude of the experimental effect, we report Cohen's d effect size for paired samples, formulated as:

$$d = \frac{\bar{d}}{s_d} \quad (4)$$

We also report the 95% confidence interval for the mean difference. Pairing by seed controls for initialisation variation and increases sensitivity to architectural effects. This protocol follows guidance on accounting for variance in machine learning benchmarks [22].

IV. RESULTS

A. Overall Performance

Table II summarises the performance of the three architectures on the test split. YOLOv8s achieves the highest mAP@0.5 together with the best inference speed. YOLO11s is comparable in accuracy but does not exceed the baseline. YOLOv8s+CBAM records the lowest mAP@0.5 of the three models.

B. Effect of Attention on Accuracy

Fig. 1 visualises the mAP@0.5 of the three models with error bars showing the standard deviation across seeds. The hypothesis that YOLO11s would outperform YOLOv8s is not confirmed by the data.

Table III reports the paired t-tests for both comparisons. YOLO11s does not differ significantly from the baseline ($\Delta = -0.0023$, $p = 0.54$). The Cohen's d of 0.61 indicates a medium effect size by convention, but the design with $n = 3$ seeds lacks the statistical power to reach significance. This result should be interpreted as an absence of evidence of a difference, not as evidence of equivalence. In contrast, adding the CBAM module produces a reduction in mAP@0.5 that is both statistically significant and practically large ($\Delta = -0.0113$, $p = 0.008$, Cohen's $d \approx 9.1$). The very small standard deviation of the CBAM variant (± 0.0004) confirms that this reduction is stable across all tested initialisations.

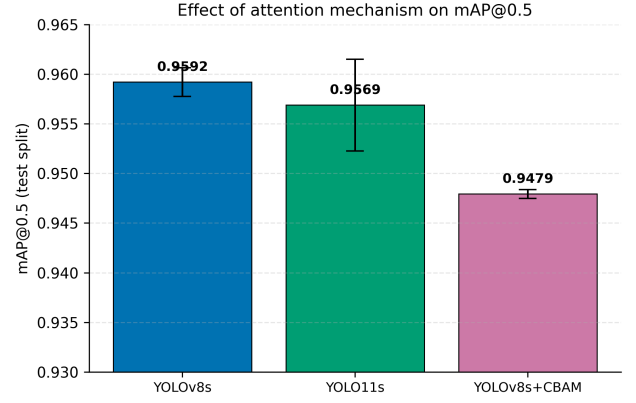


Fig. 1. mAP@0.5 on the test split across the three architectures. Error bars show the standard deviation across seeds.

TABLE III
PAIRED T-TEST RESULTS FOR MAP@0.5 (BASELINE VS EACH VARIANT).

Comparison	Δ mAP@0.5	$t(df)$	p	Cohen's d
vs YOLO11s	-0.0023	0.73 (2)	0.54	0.61
vs CBAM	-0.0113	10.9 (2)	0.008	9.1

C. Per-Class Performance

On the YOLOv8s baseline, per-class accuracy follows a pattern driven by object size and visual distinctiveness, as shown in Fig. 2. The motorcyclist class is the easiest to detect because riders are large and prominent in the frame. The helmet class is the most challenging because helmets are small, visually similar to other head accessories, and highly variable in colour and viewing angle. Because the helmet class is the most application-critical, it is the primary determinant of remaining room for improvement.

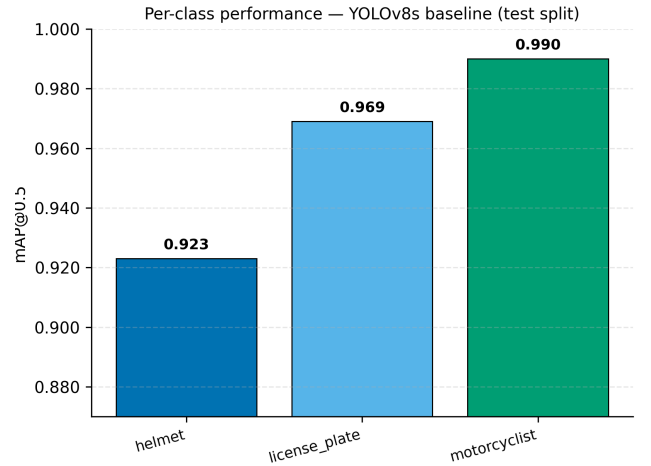


Fig. 2. Per-class performance of YOLOv8s on the test split. The helmet class is the most challenging due to its small size and visual ambiguity.

D. Speed Trade-Off

As shown in Fig. 3, YOLO11s runs at approximately 189 FPS, slower than the YOLOv8s baseline at approximately 296 FPS. The CBAM variant is nearly as fast as the baseline because the added modules are lightweight. Given that YOLO11s offers no statistically significant accuracy gain over YOLOv8s, this speed reduction is an unrewarded cost. YOLOv8s therefore occupies the most favourable position, leading on both accuracy and speed simultaneously.

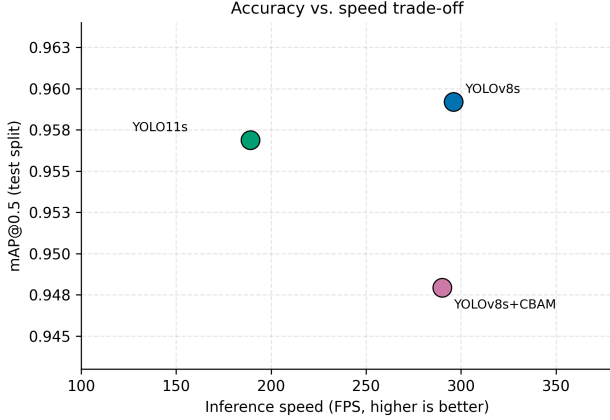


Fig. 3. Accuracy (mAP@0.5) versus speed (FPS). YOLOv8s occupies the most advantageous position because it leads on both axes.

E. Qualitative Analysis

Fig. 4 shows a typical detection output from YOLOv8s on a test image. The rider, helmet, and license plate are detected simultaneously with tight bounding boxes. Visual inspection of the test sample confirms that remaining errors concentrate on very small or partially occluded helmets, which is consistent with the per-class quantitative findings.



Fig. 4. Example detection output from YOLOv8s on a test image. All three classes are detected simultaneously.

V. DISCUSSION

The hypothesis that YOLO11s would outperform YOLOv8s was not confirmed. Under a unified protocol on this dataset, no attention variant surpassed the plain CNN, and the CBAM

module was significantly detrimental. Three explanations account for this pattern.

Dataset characteristics and ceiling effect. The training set contains 1,563 images, and the baseline already reaches approximately 0.96 mAP@0.5. With so little headroom, a more complex mechanism has almost no room to demonstrate an advantage. Attention mechanisms benefit most from large-scale data where they can learn meaningful long-range dependencies [16], [17]. In the small-data regime, the theoretical advantage of attention does not materialise.

Initialisation disruption for CBAM. The CBAM modules are randomly initialised while the rest of the network inherits COCO weights. The fresh modules disrupt the mature feature flow obtained from transfer learning. With limited data, there is insufficient gradient signal to recover from this disruption, resulting in a small but stable reduction in mAP@0.5. This effect is consistent across all three seeds, as evidenced by the very small standard deviation of ± 0.0004 .

Rival hypothesis for the CBAM reduction. The CBAM reduction does not imply that CBAM is intrinsically ineffective for helmet detection. Several design factors may explain it. The module placement at the detection head (P3, P4, P5) rather than in the backbone may be suboptimal. Training with the same learning rate for both the pretrained and the freshly initialised parameters may not give the attention layers sufficient time to converge. A systematic ablation that varies placement, lengthens training, and applies a separate learning rate for the attention parameters is necessary to isolate the exact cause. This remains part of future work.

Scope of findings. This study does not conclude that attention is universally unhelpful. The findings apply specifically to a small, relatively clean dataset and a training protocol optimised for YOLO. Under more challenging conditions, such as nighttime CCTV footage with heavy occlusion and high traffic density, the capacity of attention to model global context may prove beneficial. The practical implication within this scope is that YOLOv8s remains the most rational choice for real-time helmet detection on similar data.

VI. THREATS TO VALIDITY

Single small dataset. The dataset is curated, well-lit, and lightly occluded. These conditions differ substantially from real road CCTV. Generalisation to more challenging data, where attention mechanisms may genuinely help, has not been verified.

Limited seeds and statistical power. With $n = 3$ seeds and a baseline near the ceiling at 0.96 mAP@0.5, the paired t-test has low power to detect small improvements. The equivalence of YOLO11s and the baseline should therefore be read as an absence of evidence of a difference. Increasing to at least five seeds per model would strengthen any future equivalence or superiority claim.

Unified hyperparameters. Hyperparameters were not tuned per architecture. A protocol tuned for YOLO may disadvantage YOLO11s or CBAM-augmented models that

require different learning rates, warm-up schedules, or augmentation recipes to reach their potential. The reported results for attention-based models may represent lower bounds rather than their true capability.

CBAM placement and design. The CBAM modules were inserted only at the detection head. Placement in the backbone, training from scratch, or using a separate learning rate for the attention parameters might yield different results. Without a systematic ablation, the observed reduction cannot be attributed solely to the attention module itself.

VII. CONCLUSION

We compared three YOLO-family architectures (YOLOv8s, YOLO11s, YOLOv8s+CBAM) for motorcyclist helmet-use detection under an identical training protocol with multi-seed evaluation and paired t-tests. The hypothesis that YOLO11s would outperform the baseline was not confirmed. YOLOv8s achieved the highest mAP@0.5 (0.9592 ± 0.0014) and the fastest inference speed (~ 296 FPS). YOLO11s did not differ significantly from the baseline ($p = 0.54$), and the CBAM module significantly reduced accuracy ($p = 0.008$, Cohen's $d \approx 9.1$). These findings serve as a practical caution against assuming that newer or more complex architectures automatically outperform simpler baselines on small-scale helmet-detection tasks.

Future work should test generalisation on larger and more challenging datasets, including real CCTV footage. Per-architecture hyperparameter tuning would establish whether the equivalence of YOLO11s reflects a genuine parity or a suboptimal training recipe. A systematic CBAM ablation varying placement, training duration, and learning rate allocation would clarify whether the observed reduction is inherent or an artefact of the experimental design.

STATEMENTS

Data Availability. The dataset is publicly available through Roboflow Universe under the name NCKH 2023 Helmet Detection Project, version 19, with the MIT license. The training scripts, configurations, and experiment notebooks are available in the authors' research repository.

Conflict of Interest. The authors declare no conflict of interest.

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